Indian Institute of Technology, Madras

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MILESTONE 1 REPORT

Civic Services Platform

Pothole Reporting and Resolution System

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1. Introduction

> *The Civic Services Platform aims to bridge the communication gap between citizens and municipal authorities by providing a centralized, transparent, and accountable system for pothole reporting and resolution.*

Road potholes are one of the major causes of road accidents, traffic congestion, and vehicle damage. Citizens usually report potholes through phone calls, emails, social media posts, WhatsApp messages, or physical complaints. However, these methods often suffer from a lack of transparency, the absence of complaint tracking, duplicate reports, and delays in maintenance activities.

The platform enables citizens to submit complaints along with photographs and location details, while municipal authorities can efficiently assign repair work and monitor progress. Field workers can update repair status and upload proof of completion, thereby improving accountability and transparency.

1.1 Problem Statement

The existing process for reporting and resolving potholes is fragmented and inefficient. Citizens are unable to track the status of their complaints, administrators face difficulties managing complaints received from multiple channels, and field workers lack a structured mechanism to record and update maintenance activities. This often leads to delays, duplicate complaints, poor coordination, and a lack of accountability.

1.2 Objectives

The objectives of the proposed system are:

- To provide a centralized platform for reporting potholes.
- To enable citizens to track the status of their complaints.
- To improve transparency and accountability in complaint handling.
- To facilitate the assignment and monitoring of repair work.
- To maintain historical records of complaints and their resolutions.
- To provide reporting and analytics capabilities for better decision-making.

1.3 Scope of the Project

The proposed system focuses on managing the complete lifecycle of pothole complaints — from complaint registration to repair completion and feedback collection. The system will support:

- Registration and authentication of users.
- Reporting potholes with descriptions and photographs.
- Tracking complaint status in real time.
- Assignment of complaints to field workers.
- Updating repair progress and uploading proof of completed work.
- Collecting citizen feedback after resolution.
- Generating reports and statistics for management decision-making.

The system is intended for Citizens, Field Workers/Engineers, and Administrators. The Administrator is a consolidated role responsible for complaint management, monitoring, reporting, user management, and system administration.

2. Identification of Users

The stakeholders are classified into primary, secondary, and tertiary users based on their frequency and nature of interaction with the system.

2.1 Primary Users — Citizens

Citizens are the direct beneficiaries of the platform. They report potholes, track complaint status, verify repairs, and provide feedback after issue resolution.

Responsibilities:

- Report potholes with photographs and location details.
- View and track complaint history and status.
- Verify completed repairs and provide feedback/ratings.

2.2 Secondary Users — Field Workers / Engineers

Field workers or engineers are responsible for executing repair activities and updating the progress of assigned work.

Responsibilities:

- View and access details of assigned complaints.
- Perform repair activities and update progress status.
- Upload proof of completed work and maintain records.

2.3 Tertiary Users — Administrator

The Administrator is a consolidated role combining municipal administration, senior management, and system administration responsibilities. This role oversees the overall functioning of the platform — supporting strategic decision-making, operational monitoring, and platform administration.

Responsibilities:

- Review, prioritize, and manage all reported pothole complaints.
- Assign repair tasks to field workers or maintenance teams.
- Monitor complaint resolution progress and maintain historical records.
- Generate reports, statistics, and performance summaries.
- Analyze complaint trends to identify high-risk areas.
- Manage user accounts, roles, and access permissions.
- Ensure system security, availability, data backup, and recovery.
- Support informed decision-making through operational insights and analytics.

2.4 Summary of Users

User Type	Category	Key Responsibilities
Citizen	Primary User	Report potholes, track complaints, provide feedback
Field Worker / Engineer	Secondary User	Repair potholes and update status
Administrator	Tertiary User	Supervise complaint management, assign work, monitor progress, generate reports, manage users, and maintain the overall system

3. Interview Findings and Pain Points

Interviews and observation studies were conducted with all three stakeholder groups. The findings reveal several inefficiencies in the existing pothole reporting and maintenance process, providing the foundation for user requirement identification and user story creation.

3.1 Citizen Interviews

Most citizens reported that the current process is inconvenient and lacks transparency. Complaints are submitted through informal channels with no acknowledgement or status updates.

ID	Pain Point Description
P1	Lack of a Centralized Reporting Mechanism: Citizens use multiple channels to report road damage, resulting in inconsistency and confusion.
P2	No Complaint Tracking Facility: After reporting an issue, citizens cannot determine whether their complaint has been registered or any action initiated.
P3	Lack of Transparency: Citizens are not informed about the progress or completion of repair activities.
P4	Delayed Resolution: Complaints often remain unresolved for long periods, causing inconvenience and safety concerns.
P5	Duplicate Reporting: Since there is no visibility into previously reported issues, multiple citizens may report the same pothole repeatedly.
P6	Lack of Accountability: Citizens are unable to verify whether repair work has actually been completed.
P7	Absence of Feedback Mechanism: Citizens have no formal mechanism to express satisfaction or dissatisfaction with the services provided.

3.2 Administrator Interviews

Administrators stated that handling complaints from multiple sources is difficult and time-consuming. Monitoring and coordinating repair activities pose significant operational challenges.

ID	Pain Point Description
P8	Complaints Arrive Through Multiple Channels: Managing complaints from phone calls, emails, WhatsApp, and social media becomes complicated.
P9	Difficulty in Prioritization: Determining which complaints require immediate attention is challenging without a structured system.
P10	Manual Assignment Process: Assigning tasks to field workers is performed manually, increasing the possibility of errors and delays.

P11	Limited Visibility of Complaint Status: Administrators find it difficult to monitor the progress of ongoing repair activities.
P12	Duplicate Complaints: Multiple reports regarding the same pothole increase workload and complicate complaint management.
P13	Difficulty in Maintaining Historical Records: Tracking completed complaints and maintaining records for future reference is cumbersome.
P14	Lack of Performance Monitoring: Generating statistics and evaluating maintenance efficiency is difficult without a centralized system.

3.3 Field Worker / Engineer Interviews

Field workers reported several operational difficulties during execution of repair activities, primarily arising from poor communication and lack of a structured assignment system.

ID	Pain Point Description
P15	Unstructured Assignment Process: Work assignments are often communicated verbally or through informal messages.
P16	Insufficient Information: Field workers sometimes receive incomplete information regarding the location and severity of potholes.
P17	Difficulty in Updating Progress: There is no structured mechanism for reporting the progress of repair activities.
P18	Lack of Proof of Completion: Field workers have difficulty maintaining evidence of completed work.
P19	Poor Record Maintenance: Historical records of completed assignments are not maintained systematically.
P20	Communication Gaps: Communication between administrators and field workers is often inefficient, resulting in delays and misunderstandings.

3.4 Summary of Pain Points

Stakeholder	Major Pain Points
Citizens	No centralized reporting, lack of tracking, delayed resolution, lack of transparency
Administrators	Multiple complaint channels, manual assignment, duplicate complaints, poor monitoring
Field Workers	Unstructured assignments, insufficient information, lack of progress tracking
Overall System	Lack of accountability, poor communication, absence of historical records

4. User Stories

User stories were developed based on requirements gathered through interviews and observation studies, following SMART guidelines. They capture the expected functionality and value delivered by the system from each stakeholder's perspective.

4.1 User Stories for Citizens

US1	As a **citizen**, *I want to* create an account, *so that* I can securely access the platform and manage my complaints.
US2	As a **citizen**, *I want to* log into the system, *so that* I can access my complaint history and track their status.
US3	As a **citizen**, *I want to* report a pothole, *so that* the authorities become aware of road damage requiring maintenance.
US4	As a **citizen**, *I want to* provide a description of the issue, *so that* maintenance personnel can understand the nature of the problem.
US5	As a **citizen**, *I want to* upload photographs of potholes, *so that* authorities can assess the severity of road damage.
US6	As a **citizen**, *I want to* specify the location of the pothole, *so that* repair teams can reach the correct location.

US7	As a **citizen**, *I want to* view my submitted complaints, *so that* I can keep track of issues reported by me.
US8	As a **citizen**, *I want to* track the status of my complaints, *so that* I know whether any action has been taken.
US9	As a **citizen**, *I want to* verify completed repairs, *so that* I can ensure that the issue has been resolved.
US10	As a **citizen**, *I want to* provide feedback after issue resolution, *so that* I can express my level of satisfaction with the service.

4.2 User Stories for Administrators

US11	As an **administrator**, *I want to* view all reported complaints, *so that* I can manage civic issues efficiently.
US12	As an **administrator**, *I want to* prioritize complaints, *so that* severe road damages are addressed first.
US13	As an **administrator**, *I want to* assign complaints to field workers, *so that* repair activities can be initiated promptly.
US14	As an **administrator**, *I want to* monitor complaint status, *so that* I can track the progress of maintenance activities.

US15	As an **administrator**, *I want to* identify duplicate complaints, *so that* unnecessary effort and duplication of work can be avoided.
US16	As an **administrator**, *I want to* maintain historical records, *so that* completed work can be referred to in the future.
US17	As an **administrator**, *I want to* generate reports and statistics, *so that* I can evaluate departmental performance.
US18	As an **administrator**, *I want to* view complaint statistics, *so that* I can understand the overall condition of road infrastructure.
US19	As an **administrator**, *I want to* analyze complaint trends, *so that* areas with recurring problems can be identified.
US20	As an **administrator**, *I want to* monitor the performance of maintenance operations, *so that* resources can be utilized effectively.
US21	As an **administrator**, *I want to* access reports and analytics, *so that* I can make informed decisions.
US22	As an **administrator**, *I want to* manage user accounts, *so that* authorized users can access the platform securely.
US23	As an **administrator**, *I want to* assign roles and permissions, *so that* access to system resources can be controlled.

US24	As an **administrator**, *I want to* monitor system activities, *so that* abnormal behavior can be detected and investigated.
US25	As an **administrator**, *I want to* maintain backup and recovery mechanisms, *so that* important information can be restored in case of failures.
US26	As an **administrator**, *I want to* deactivate inactive user accounts, *so that* unauthorized access can be prevented.

4.3 User Stories for Field Workers / Engineers

US27	As a **field worker**, *I want to* view tasks assigned to me, *so that* I know which potholes require repair.
US28	As a **field worker**, *I want to* access complaint details and locations, *so that* I can reach the affected area efficiently.
US29	As a **field worker**, *I want to* update the progress of repair work, *so that* administrators can monitor ongoing activities.
US30	As a **field worker**, *I want to* mark tasks as completed, *so that* complaint resolution can be recorded.
US31	As a **field worker**, *I want to* upload evidence of completed work, *so that* completed repairs can be verified.

US32

As a **field worker**,
I want to maintain records of completed assignments,
so that I can review previous work when necessary.

4.4 Summary

A total of thirty-two user stories were identified across all three stakeholder groups. These user stories will serve as the foundation for product backlog creation, system design, sprint planning, and implementation activities in subsequent project phases.

Stakeholder	User Story IDs	Count
Citizens	US1 – US10	10
Administrators	US11 – US26	16
Field Workers / Engineers	US27 – US32	6
Total	—	32